

Introduction to Nephrology

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Introduction to Nephrology / Investigations of the Renal System

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Outline

Introduction · Basic anatomy of the kidneys · Functions of the kidney · Common renal disorders · Features of renal disorders · Investigations in nephrology · Conclusion · References

Introduction

Nephrology is the branch of paediatrics responsible for the **diagnosis, evaluation and management of the disorders of the kidneys and urinary tract in children.**

It focuses on **kidney health and diseases**, and covers a wide range of disorders including **glomerular and urinary tract diseases** – glomerulonephritis, AKI, CKD, nephrotic syndrome, hypertension, acid-base disorders, CAKUT, UTI etc.

A nephrologist should be able to

- Evaluate and diagnose renal pathology
- Institute appropriate treatment
- **Monitor growth and development** in children with kidney disorders
- Provide **renal replacement therapy**
- **Educate and counsel families** on reno-prevention and long term care

Basic anatomy

Basic anatomy of the kidneys

- **Paired, bean-shaped, retroperitoneal organs** located between **T12-L3** vertebrae, with the **left kidney being more superior than the right**
- Covered by the **renal fascia – Gerota’s fascia**
- Weigh about **150 g in males and 135 g in females**

- About **10-12 cm** in length, **5-7 cm** in width, **2-3 cm** in thickness

Blood supply and innervation

- Receives about **20% of cardiac output**, from the paired **renal arteries at the level of L2**
- **Venous drainage** by the renal veins
- **Lymphatic drainage** via renal lymphatics
- Receives **both sympathetic and parasympathetic** innervation:
 - **Sympathetic** arises from the spinal cord at **T8-L1**
 - **Parasympathetic** comes from the **vagus nerves**
 - These cause **vasoconstriction and vasodilation** respectively

Structure

The nephron

Divided into the **cortex** and **medulla**. The **functional unit is the nephron**, made of:

- **The renal corpuscle** – glomerulus and Bowman's capsule
- **PCT** – located in the renal cortex
- **Descending loop of Henle** – from cortex to medulla
- **Ascending limb**
- **Distal convoluted tubule** – cortex
- **Collecting duct** – from cortex, passes through medulla to renal pelvis

Functions of the kidneys

1. **Filtration and excretion of metabolic waste products**
2. **Fluid and electrolyte balance**
3. **Blood pressure regulation**
4. **Endocrine function**
5. **Acid-base homeostasis**
6. **Regulation of red blood cell production**
7. **Calcium and bone metabolism**

Common renal disorders

- **CAKUT**
- **UTI**
- **Nephrotic syndrome**
- **AGN**
- **AKI**
- **CKD**
- **Vesicoureteric reflux (VUR)**
- **Haemolytic uraemic syndrome**
- **Renal tubular disorders**

Risk factors for renal disorders

- **Drug toxicity**
- **Family history / genetics**
- **Obesity**

- Hypertension
- Prematurity
- Diabetes
- Autoimmune disorders

Clinical features of renal disorders

- Oliguria / anuria
- Oedema
- Haematuria
- Proteinuria
- Hypertension
- Electrolyte imbalance
- Lower abdominal pain

Investigations in nephrology

Rationale

- To confirm a clinical diagnosis
- To assess structural and functional integrity of the kidneys/urinary tract
- To assess the severity and extent of the disease
- Prognostication
- To exclude other co-morbidities

Consideration to diagnostic value, availability, cost and sustainability is very important.

Classification of investigations

1. Urine examinations
2. Blood investigations
3. GFR estimation
4. Assessment of tubular function
5. Imaging
6. Biopsy
7. Bladder function assessment

Urine examination

An important **non-invasive** test.

Diagnostic yield depends on method of collection, preparation and testing.

Samples should be **examined freshly** to avoid **disintegration of cellular elements and bacterial overgrowth**. Urine can be **stored in the refrigerator at 4 °C**.

Urine examination

Methods of urine collection

- MSU (with a long bladder time)
- Clean catch method

- **Bagging** – associated with **high risk of contamination**; the bag must be removed as soon as urine is passed. *Important for its negative predictive value*
- **SPA (suprapubic aspiration)** – **most reliable**, useful in children **less than 2 years**
- **Catheterization**
- **Use of mid-stream urinary collector**

Interpretation of culture

Interpretation of the culture report depends on the method of urine collection.

- **SPA** – **any growth is significant**, because bladder urine is supposed to be sterile
- **CSU** – **≥ 10 CFU/mL** considered significant
- **MSU** – **≥ 10 CFU/mL** considered significant (a second or repeat urine sample may be required for diagnosis, particularly if there are no referable symptoms and signs to the urinary tract)

Method	Finding
Suprapubic aspiration	If a UTI is present, bacteria are likely to be proliferating in bladder urine, with growth of any organism except 2,000-3,000 CFU/mL coagulase-negative staphylococci
Catheterization in a girl, or midstream clean-void in a circumcised boy	Febrile infants and children with UTI usually have $> 50,000$ CFU/mL of a single urinary pathogen; however UTI may be present with 10,000-50,000 CFU/mL of a single organism
Midstream clean-void in a girl or uncircumcised boy	UTI is indicated when $> 100,000$ CFU/mL of a single urinary pathogen is present in a symptomatic patient. Pyuria usually present. A UTI may be present with 10,000-50,000 CFU/mL of a single bacterium
Any method in a girl or boy	If the patient is asymptomatic , bacterial growth is usually $> 100,000$ CFU/mL of the same organism on different days. If pyuria is absent, this probably indicates colonization rather than infection

Patients with **urinary frequency** (i.e. decreased bladder incubation time) are those most likely to have bacteria proliferating in the urinary bladder in the presence of **low colony counts**.

Routine dipstick

Leukocyte · Nitrite · Protein · pH · Blood · Specific gravity · Ketone · Bilirubin · Glucose · Ascorbic acid · Urobilinogen

Principle: based on the **change in colour in the respective pads** when urine constituents react with a chemical impregnated in the specific pad on the strip. The **intensity of colour gives a qualitative grading** which is read off the chart on the container.

Quality of result depends on strict adherence to the manufacturer's instructions.

Other urine examinations

Urine macroscopy – appearance (pale to dark yellow and amber), any colour change, odour

Urine microscopy – RBC, WBC, cast, crystal

Urine culture and sensitivity

Spot urine protein creatinine ratio

Estimate of urine protein concentration

Dipstick	Concentration
Trace	< 30 mg/dL
+	30 mg/dL
2+	100 mg/dL
3+	300 mg/dL
4+	> 1000 mg/dL

False results in proteinuria

False **NEGATIVE** proteinuria

- **Dilute** or **very acidic** urine
- Proteinuria from other proteins

False **POSITIVE** proteinuria

- **Concentrated** urine
- **Alkaline** urine
- **Gross haematuria**
- **Delay in reading**
- **Urine contamination** with vaginal secretions or antiseptic

Blood tests

- **Se/U/Cr**, calcium, phosphate, magnesium, **vitamin D assay**
- **FBC**, **MP**, **ESR**
- **Arterial blood gases**
- **Immunologic tests**:
 - **C3**, **C4**, **CH50**
 - **ASO titre**
 - **Anti-DNAse B**
 - **IgG**, **IgA** – **SLE**, **JRA**
 - **ANCA** – **Wegener granulomatosis**, **crescentic glomerulonephritis**
 - **Hormonal profile** – **T3**, **T4**, **TSH**

Estimation of GFR

Purpose: to assess renal function, detection of renal disease, severity assessment, and prognosis.

There is no perfect substance for its measurement.

Commonly used markers

Inulin · polyfructose · Cr-EDTA · creatinine · cystatin C · urea

The ideal marker should

- Be freely filtered at the glomerulus
- Be able to achieve a stable plasma concentration
- Not be reabsorbed, secreted or metabolized by the kidney

Why creatinine is not ideal

- It is secreted in the proximal tubule
- Its concentration is influenced by age, sex, height and dietary protein

Creatinine clearance is the most widely used, done using a timed urine collection (24 or 6-12 hours).

Why urea is not ideal

- It is reabsorbed by the renal tubules
- Plasma level is affected by dietary protein intake and state of hydration

The Schwartz formula

$$\text{GFR (mL/min/1.73 m}^2\text{)} = 0.413 \times \text{height (cm)} / \text{serum creatinine (mg/dL)}$$

GFR is expressed per 1.73 m² – the average body surface area of an ideal adult.

Assessment of renal tubular function

Fractional excretion of electrolytes/solutes

The ratio of the urine concentration of the solute x plasma concentration, divided by the ratio of urine creatinine concentration to plasma creatinine concentration.

$$\text{FENa (\%)} = \text{quantity of Na excreted} / \text{quantity of Na filtered} \times 100 = (\text{UNa} \times \text{Pcr}) / (\text{PNa} \times \text{Ucr}) \times 100$$

Extracellular fluid volume normally determines the urinary Na excretion.

- FENa ≤ 1 in normovolaemia and pre-renal AKI
- FENa > 1 in impaired tubular transport, e.g. ATN

Other tubular function tests

Concentrating ability test – osmolality, pH

Water deprivation test – to differentiate between central DI and nephrogenic DI

Diluting ability test – for assessment of unexplained hyponatraemia

Urine acidification test – serial urine pH, bicarbonate excretion, titratable acid, NH₄⁺ excretion, and ammonium chloride challenge test

Renal imaging

Ultrasonography (abdominal USS KUB)

Ultrasonography makes use of sound waves beyond the human auditory capability (> 20 kHz).

- Sound wave is **produced by the transducer**, and the **reflected sound wave is also received by the transducer**
- The received echoes are **converted to electrical impulses** by the transducer
- The electrical impulses are subsequently **displayed as images on the monitor**

Has become a routine imaging modality. Varied uses:

- Investigating **causes of urinary tract infections**
- Checking for **causes of urinary tract obstruction**
- Diagnosing **kidney stones / masses / cysts / tumours**
- Evaluating **congenital urinary tract abnormality**
- Assessing **kidney injury from trauma**
- Monitoring **kidney transplant status** or vascular issues

Merits of ultrasonography:

- **No ionizing radiation**
- **Real time images**
- **Readily available**
- **Relatively cheap**
- **Portable, point-of-care option**

Demerits of ultrasonography:

- **Cannot see through compact bone** (posterior acoustic shadow)
- **Gas distorts image**
- **User dependence**
- **Depth limitation**

Radiographic imaging

Plain film · intravenous urography · voiding cysto-urethrography · antegrade and retrograde pyelography · renal arteriography

Plain film Plain film

Plain film

- For diagnosis of **nephrocalcinosis, radio-opaque stones** in pelvis, ureters and bladder
- Diagnosis of **congenital spinal abnormalities** with urinary symptom manifestation
- **Before injecting contrast material** for intravenous urography
- **Declining with the advent of USS**

Intravenous urography (IVU) Intravenous urography

- To **demonstrate anatomy of the urinary tract**, as it can give details of the **calyces and ureters**

- In the diagnosis of **occult duplex kidney, malrotation, site of urinary tract obstruction and calculi**
- Can **detect renal scars**
- **Demerits:** high radiation exposure, low sensitivity for renal scar detection, potential reaction to contrast; **not used in neonates, young infants and patients with renal failure**

Voiding cysto-urethrography (VCUG) Voiding cysto-urethrography

- **Definitive method for demonstrating the lower urinary tract**
- **Gold standard for diagnosing and grading VUR and PUV obstruction**
- Makes use of **contrast injected into the bladder** to the upper tract
- **Films are taken during instillation, voiding and post-voiding**

Indications for VCUG:

- Suspected ureteric dilatation
- Small contracted kidneys
- Renal failure of indeterminate cause
- Terminal haematuria
- Voiding difficulties
- Thick walled bladder seen by US

Demerits: high dose of radiation, risk of UTI

Radioisotope studies

Dynamic renography – uses radioisotopes like **technetium-labelled diethylene-triamine penta-acetic acid (Tc-DTPA)** and **Tc-mercapto-acetyl-triglycine (Tc-MAG3)**

Uses:

- Determining **differential renal function**
- **Indirect voiding cystography**
- Evaluation of **post-transplantation state**
- Evaluating **surgical treatment** involving the kidneys

Has low dose of radiation. Demerits: poor visualization of renal cortex, and cannot be used to evaluate renal size or morphology.

Static renal scan:

- In **renovascular diseases** and **dribbling urinary incontinence**
- **Location of ectopic/absent kidney**
- **Detection of renal scars**
- **Acute phase of pyelonephritis** – to demonstrate parenchymal involvement
- To detect **focal parenchymal abnormalities**

Drawback: time-consuming; does not give information on function of the urinary tract.

Computerized tomography (CT)

- **Indications similar to that of USS**
- Gives **clearer resolution**
- Can detect **fine calcifications** within the renal parenchyma and collecting system

- **With contrast enhancement**, can evaluate masses in the pelvis or bladder, e.g. Wilms tumour
- **Expensive**; risk of exposure to considerable radiation

Magnetic resonance imaging (MRI)

- For **detailed evaluation of complex urinary system disorders**
- Gives **excellent pelvic floor details**
- Good for **renal masses and cystic lesions**
- **No radiation exposure**
- **Heavy sedation needed**

Renal biopsy

Renal biopsy is the process of **obtaining tissue from the kidney in an individual for the purpose of histopathological evaluation.**

Types

- **Percutaneous**
- Transvenous or transjugular
- Laparoscopic
- Open surgical approach

The percutaneous approach is preferred by the paediatric nephrologist.

Renal biopsy needles

Renal biopsy needles

Renal biopsy needles

Renal biopsy needles

- **Vim-Silverman needle**
- **Manual Tru-cut biopsy needle**
- **Semi-automatic Tru-cut biopsy needle**

Tru-cut spring-loaded automatic biopsy system

Tru-cut spring-loaded automatic biopsy system

- **Tru-cut spring-loaded automatic biopsy system**

General indications for renal biopsy

- **Establishment of the exact diagnosis**
- As an aid to **determine the nature of recommended therapy**
- To help **decide when treatment is futile**
- To ascertain the **degree of active (potentially reversible) and chronic changes (irreversible)**
- **Follow up** of treatment or disease
- **Research**

Prognostication based on kidney pathology alone may be affected by sample size, especially in focal lesions – biopsies with glomeruli < 5 may not be accurate.

Indications by condition

Nephrotic syndrome

- Age < 1 or > 7 years
- **SRNS** (steroid-resistant nephrotic syndrome)
- Those with **evidence of nephritis** – hypertension, haematuria, low C3, or decreased renal function persisting despite volume correction

Acute glomerulonephritis

- **Course not typical of PSAGN**
- **HSP or SLE**
- Assessment of **severity of injury**, to guide therapy and prognosis
- **Differentiation of specific types of proliferative lesions** e.g. MPGN, C3 glomerulopathies including dense deposit disease and C3-dominant GN
- **Recurrent persistent haematuria, proteinuria**

AKI

- **Unexplained AKI** – when pre- and post-renal causes are excluded
- AKI associated with **nephritis, nephrotic syndrome, evidence of vasculitis or systemic disease** – biopsy usually performed
- **Persisting ATN** when the cause remains uncertain after complete evaluation
- **Suspected RPGN** – determination of cause, activity and chronicity
 - **Serology:** anti-GBM antibody, ANCA – screening test for necrotizing vasculitides
 - **Immunofluorescence:** p-ANCA, c-ANCA, MPO or PR3

CKD

- **When kidneys are not shrunken**
- Diagnosis of primary disease
- Assessment of severity of morphologic lesions
- Determination of **risk of recurrence in eventual renal transplant**
- Suitability of **deceased versus living-related donor transplantation**

Systemic diseases

- To assess **severity of renal disease** e.g. SLE, atypical HUS
- Patients with **diabetes with a clinical course atypical of diabetic nephropathy**

Renal transplant

- **Acute rejection**
- **Recurrence of kidney disease**
- **Calcineurin toxicity**
- Some infections
- **Chronic allograft rejection**

Conditions NOT requiring renal biopsy

- **Isolated glomerular haematuria (microscopic)**

- Isolated non-nephrotic haematuria
- Childhood nephrotic syndrome age > 1 yr and < 7 yrs
- Frequently relapsing nephrotic syndrome

Analysis – specimen transport medium

Examination	Transport medium
Light microscopy	10% buffered formalin
Immunofluorescence	Michel's transport media
Electron microscopy	Glutaraldehyde

Contraindications to renal biopsy

- Uncorrectable bleeding diathesis – **ABSOLUTE** contraindication
- Uncontrolled hypertension
- Active renal or peri-renal infection
- Hydronephrosis
- Obesity
- Ascites
- Small shrunken kidneys
- Tumours, large cysts, abscesses, or pyelonephritis
- Solitary, ectopic, or horseshoe kidney
- Uncooperative patient
- Skin infection over the biopsy site
- When a skilled operator or appropriate pathology support is not available

Patient preparation

- Patient should be **admitted at least the day before** the procedure
- **History** – bleeding disorders
- Ensure patient is **not on warfarin, any anticoagulant or antithrombotic agent**, and has had **no aspirin or other NSAID for at least 7 days**
- **Check blood pressure** – hypertension should be **controlled before biopsy**
- Patient/parents to be **informed of the procedure, risks explained, and consent obtained**

Investigations before biopsy:

Test	Requirement
PCV	> 24-30%
FBC – platelet count	at least 100,000/mm³
PT	Coagulation profile
INR	< 1.2
APTT	< 2x control or < 40 secs
E&U and creatinine	
Abdominal USS KUB	to assess if patient has two kidneys , kidney sizes etc
Urinalysis, urine m/c/s	
Group and cross-match	1 unit of blood

Further preparation:

- If the child is on **haemodialysis**, the procedure should be done **after at least 24 hours** of the last dialysis session
- For patients with **prolonged bleeding time (> 8-10 minutes**, e.g. in SLE, azotaemia), **IV desmopressin 0.3 µg/kg 30 min prior**, or **intranasal DDAVP 2-4 µg/kg 2 hours before** the procedure, may be considered

Premedication

- **NPO 4 hours before** the procedure, and commence **5% D/W at maintenance**
- **Local anaesthesia and sedation**
- **Sedation** – most widely used: **1-2 doses of midazolam (0.1 mg/kg)** and **ketamine 0.5-1.0 mg/kg** (with anaesthetist)
- **± IV atropine 0.01 mg/kg**, 1-2 minutes after midazolam

Procedure

Persons in attendance: paediatric nephrologist, radiologist, pathologist (presence of radiologist or pathologist may vary across centres)

Position:

- **Prone position** for biopsy of **native kidney**
- **Supine position** for **graft kidney**
- In conditions like **abdominal distension and ascites**, the biopsy can be done in the **lateral decubitus** position

During: oxygen saturation, heart rate and blood pressure monitoring

- **The LEFT kidney is usually biopsied**
- Identify the **lower pole** of the kidney under **real-time USS**, and determine the skin entry point
- Clean skin with **methylnated spirit and povidone iodine**
- Give local anaesthesia with **1% lignocaine** (under USS guidance)
- Make a **nick on the skin** at the site of entry
- **Advance the biopsy needle under real-time ultrasound guidance into the cortex** of the kidney
- **16G versus 18G needle**
- **Sample is obtained from the renal CORTEX**
- **2-3 passes**
- **The core can be examined for the presence of glomeruli**

Post-biopsy instructions

- Do a **post renal biopsy clinical evaluation**
- **Monitor vital signs** – BP, heart rate, respiratory rate, temperature: **¼ hourly for 1 hour**, **half hourly for 2 hours**, **hourly for 4 hours**, then **4 hourly for 24 hours**
- **Keep urine rack**
- **IV fluids 10% dextrose** at maintenance requirements
- **Fluid input and output monitoring**
- Patient can eat as soon as fully conscious
- **Watch out for gross haematuria, abdominal pains, or flank pains**

Complications

- Significant haematuria
- Anaemia and hypotension
- Bowel or solid organ perforation / laceration
- Loss of a kidney
- Death
- AV malformation
- Hypertension
- Persistent loin pain

Evaluation of bladder function – urodynamic studies

Urodynamic studies are investigations done to evaluate the anatomy and physiology of the lower urinary tract, especially the filling and emptying of the bladder and urethra.

Useful in:

- Neurogenic bladder
- Urinary incontinence
- Voiding dysfunction
- VUR
- Anorectal malformations

Non-invasive studies:

- Uroflow study / uroflowmetry – measures urine speed and volume
- Post-void residual urine estimation

Invasive studies:

- Cystometry studies
- Pressure flow study
- Electromyography

Conclusion

- Nephrology is an important aspect of paediatrics
- Understanding of renal physiology is invaluable in recognizing and managing renal disorders
- Various investigations and imaging are indispensable in making diagnosis, patient monitoring and prognostication

References

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